

Saarthak Saxena

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Experience

Applied AI Intern

Mar 2026 – May 2026

Bruce.so

- Data Pipeline & Optimization:** Designed and shipped a relationship-scoring strategy that reduced posts processed by **97%** (2,186 to 64 data points) while retaining **98%** of high-quality relationship signals, validated through **60+** controlled experiments across **72** profiles and **~90K** baseline relationship records. Built the quality score using recency, engagement, virality controls, top-result filtering, and reaction limits, then reduced third-party API usage by consolidating a multi-step workflow into a single pass and cutting external API calls per profile refresh by **~60%**.
- Search & Ranking:** Rebuilt people search ranking so expanded LLM queries could enforce exact role, company, and location requirements, replacing soft score cutoffs with deterministic filtering for structured searches.
- Knowledge Graph & Taxonomy:** Built an autonomous skill-taxonomy maintenance engine that extracts skills and tools from profile text, ranks evidence with semantic similarity and usage patterns, and classifies terms into parent-child categories.
- Decision Quality:** Built a review system that revisits uncertain skill matches after collecting more evidence across batches.
- Connection Path Finding:** Built a path expansion system that finds direct and indirect relationship paths up to **3** degrees away across **~4,700** profiles and **~11,600** relationship links, evaluating **3.4M+** candidate path checks and **29.3M** possible route combinations to improve candidate selection. Achieved a **99.87%** path-finding success rate with **87.69%** best-candidate selection accuracy using threshold-based selection, validated across **9** strategies and multiple profile-connectivity segments.

UG Researcher

Jan 2025 – Present

A2A Lab, IIIT Delhi

On Site

- Optimized embedded systems performance through targeted code and hardware improvements.
- Developed an intuitive UI for visualizing complex system performance metrics.

ML Engineer

Jun 2025 – Nov 2025

Stealth Startup

Contract work: Jul 2025 – Oct 2025

- Business Impact:** Built the core ML infrastructure that helped the client secure a **INR 30 Lac (~\$35k)** research grant.
- Custom Tooling:** Adapted and extended GraphGym source code to ingest custom molecular embeddings, enabling automated experimentation across **48k** data points.
- Massive Scale:** Orchestrated **1,000,000+** optimization trials on remote distributed GPUs and beat Chemprop, the industry standard.
- Rapid Delivery:** Shipped a production-ready, lightweight **2M**-parameter inference API in **8** weeks, handling the full lifecycle from data ingestion to AWS deployment using FastAPI and Docker.

Software Engineering Intern

Jan 2024 – May 2024

HeydoTech

- Used Python libraries including OpenCV and PyAutoGUI to develop **2** automation tools for internal projects.
- Collaborated with cross-functional teams of **5+** members to implement feature enhancements, improving system performance for approximately **100** users.
- Contributed to UI/UX improvements that enhanced overall user experience, measured by positive feedback from **80%** of test users.

Projects

Underwater Mysid Detection

CV, Deep Learning, PyTorch, YOLOv8 | Jan 2025 – Present

- Built a CV pipeline for near-transparent mysid detection in underwater scenes, reaching **0.61 mAP50** with augmented **38–114** image datasets and YOLOv8-seg improvements over U-Net baselines.

Detoxify: Distraction Filter

Python, JS, MLOps, Gemini API | Jan 2025 – Present

- Built a Chrome extension using Gemini AI to classify YouTube batches in **1.95s**, analyzing **8–12** videos per batch and surfacing alerts for distracting content.

Docbook

Python, GraphRAG, CLI | Apr 2025 – Present

- GraphRAG-based CLI tool for intelligent documentation context extraction.
- Converts a single URL input into a full knowledge graph using crawl4ai, with automated semantic search and retrieval.

Ghibli Image Scraper

JS, Chrome Ext API, Gemini AI | Mar 2025

- Created a Chrome extension using Gemini AI to detect and download Ghibli-style art from X.com, earning GitHub Projects Community visibility.

Bayesian Course Recommendation System

Python, PyTorch, Bayesian ML | 2025

- Achieved **0.915** AUC using Bayesian Personalized Ranking for implicit feedback.
- Implemented matrix factorization with BPR optimization and probabilistic handling for cold-start recommendations.

Education

BTech, Electronics and Communication Engineering

Nov 2022 – Present

IIIT Delhi